

Risques climatiques

Transition-risk component

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Course positioning

This syllabus describes Gauthier Vermandel’s transition-risk component of the course *Risques climatiques*. The full course distinguishes physical risks associated with natural perils from transition risks associated with economic, technological and behavioural change. This component develops the scenario and modelling tools used to analyse transition risk and their implications for firms, assets and insurance decisions.

Learning objectives

At the end of the component, students should be able to:

- distinguish forecasts, projections, scenarios and stress tests;
- explain the chain from economic activity and climate policy to emissions, concentrations, temperature and aggregate damages;
- interpret SSP, forcing and IAM pathways as conditional scenarios;
- use DICE to compare mitigation pathways and parameter uncertainty;
- discuss the implications and limitations of model-based transition scenarios for actuarial and financial risk analysis.

Schedule and content

The component is organised as five two-hour sessions. The first two mornings include a 15-minute break between sessions.

Date and time	Session	Content
Wed. 2 Sept., 08:30–10:30	1	Economic foundations of integrated assessment models: production, growth, welfare and controls.
Wed. 2 Sept., 10:45–12:45	2	Climate model: greenhouse gases, concentrations, radiative forcing and temperature dynamics.
Wed. 9 Sept., 08:30–10:30	3	Climate scenarios and integrated assessment models: SSPs, IAMs and DICE.
Wed. 9 Sept., 10:45–12:45	4	Parametric uncertainty: calibration draws, scenario ensembles and interpretation.
Wed. 16 Sept., 08:30–10:30	5	Optimal climate policy and the social cost of carbon.

Teaching format and materials

Each session combines lecture material, equations and figures with a Python/Jupyter notebook. The materials, data and source code are organised in the five **Session1–Session5** folders. Installation instructions and the canonical dependency list are provided in **README.md** and **requirements.txt**. Python 3.12 is the validated environment.

Assessment

Closed-book written examination (*examen sur table sans document*). The date, duration and detailed instructions are communicated by the programme administration. No separate assignment or submission deadline is specified in the course information currently available.

Prerequisites

Reinsurance, reserving, pricing and economics. Students should be comfortable reading basic Python code and interpreting graphs and numerical outputs.

Course information

- Full course volume: 20 hours (16 hours of lectures and 4 hours of practical work); 2 ECTS; semester S3.
- Course instructors: Geoffrey Ecoto and Gauthier Vermandel.
- Contact: gauthier@vermandel.fr.

Selected references

Nordhaus, W. D. (2017), *Revisiting the Social Cost of Carbon*; IPCC (2021), *AR6 Working Group I*; NGFS (2024), *Climate Scenarios for Central Banks and Supervisors*.